

Graph Algorithms — complete, visual, student-friendly guide

Common definitions used everywhere

- **Graph** $G = (V, E)$. V = vertices (nodes), E = edges (u, v) .
 - **Undirected** vs **directed** (edge direction matters).
 - **Weighted** vs **unweighted** (weights on edges).
 - **Adjacency list** (list per vertex) — space $O(V + E)$.
 - **Adjacency matrix** $(V \times V)$ — space $O(V^2)$, constant-time edge check.
 - **Path**: sequence of vertices connected by edges.
 - **Simple path**: no repeated vertices.
 - **Cycle**: path starting and ending at same vertex with length ≥ 1 (simple cycle: no repeated internal vertices).
 - **Connected components** (undirected): maximal vertex sets connected by paths.
 - **In-degree/Out-degree** (directed): counts of incoming/outgoing edges.
 - **Relaxation** (used in shortest paths): if $d[v] > d[u] + w(u, v)$ then update $d[v]$.
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1) Greedy Strategy — Elements of the strategy

Idea: At each step choose the locally best option (greedy choice) and never reconsider. If greedy choice leads to globally optimum solution, the problem is *greedy-solvable*.

Elements / recipe:

1. **Greedy choice property:** There exists an optimal solution that contains the local choice made.
2. **Optimal substructure:** An optimal solution to the problem contains optimal solutions to subproblems.
3. **Feasible set & objective:** Define what partial solutions are feasible and how to measure progress.
4. **Proof technique:**
 - Show any optimal solution can be transformed (without worsening objective) to one that begins with the greedy choice. Or
 - Exchange argument: swap elements to show greedy choice is safe.
5. **Counterexample awareness:** Not all problems with optimal substructure are greedy-solvable (e.g., shortest paths in graphs with negative cycles require Bellman-Ford, not a naive greedy).
6. **Implementation pattern:** sort candidates or use a priority queue; iteratively pick best candidate that keeps solution feasible.

Common greedy problems: Interval scheduling, Fractional knapsack, Huffman coding, MST (Kruskal/Prim), Dijkstra (works greedily under non-negative weights).

2) Huffman Coding — greedy example

Problem: Given character frequencies, build a binary prefix code (no codeword is prefix of another) minimizing weighted sum of code lengths = total encoded size.

Why greedy works: Huffman = repeatedly combine two least-frequent items. Proof uses exchange argument: any optimal tree has the two least-frequent letters as siblings; combine them into one node and continue — leads to optimality.

Steps (visual):

- Frequencies: A:45, B:13, C:12, D:16, E:9, F:5
 - Build min-heap of nodes by frequency.
1. Pop smallest: F(5), E(9) → combine → node X(14).
 2. Heap: C(12), B(13), X(14), D(16), A(45)
 3. Pop C(12), B(13) → combine → Y(25).
 4. Heap: X(14), D(16), Y(25), A(45)
 5. Pop X(14), D(16) → combine → Z(30).
 6. Heap: Y(25), Z(30), A(45)
 7. Pop Y(25), Z(30) → combine → W(55)
 8. Pop A(45), W(55) → root(100)

Then assign 0/1 to edges from root; code length = depth. Weighted sum minimal.

Pseudocode (typical):

```
Huffman(frequencies):
    create min-heap H of leaf nodes (char, freq)
    while H.size > 1:
        x = H.pop_min()
        y = H.pop_min()
        z = new node(freq = x.freq + y.freq, left=x, right=y)
        H.insert(z)
    return H.pop()  // root of Huffman tree
```

Complexity:

- Using binary heap: $O(n \log n)$ where $n = \text{\#symbols}$ (heap operations).
- Space: $O(n)$ for nodes.

Visual tip: draw leaves as initial nodes, repeatedly merge two smallest into a parent (draw arcs) until single tree.

3) Minimum Spanning Trees (MST)

Problem: Given a connected, undirected, weighted graph, find a subset of edges connecting all vertices with minimum total weight (no cycles) — a tree.

Properties:

- **Cut property:** For any cut of vertices ($S, V \setminus S$), the minimum weight edge crossing the cut belongs to some MST.
- **Cycle property:** For any cycle, the maximum weight edge on the cycle does not belong to any MST.

Two classic greedy MST algorithms: **Kruskal** and **Prim**.

Kruskal's Algorithm

Idea: Sort edges by weight ascending; add edges in that order if they don't form a cycle (use Disjoint Set Union/Union-Find to detect cycles).

Pseudocode:

```
Kruskal(G) :  
  A = {}  
  for each vertex v:  
    MakeSet(v)  
  sort edges E by weight ascending  
  for each edge (u,v) in sorted E:  
    if Find(u) != Find(v):  
      A.add((u,v))  
      Union(u,v)  
  return A // A is MST edges
```

Data structure: Union-Find with path compression & union by rank $\rightarrow$ nearly $O(\alpha(V))$ per op.

Complexity:

- Sorting edges: $O(E \log E)$.
- Union-Find ops: $O(E \alpha(V)) \approx O(E)$.
- Total: $O(E \log E)$ or $O(E \log V)$ since $E \leq V^2$.
- Space: $O(V + E)$.

Visualization: draw graph with numbered edges; take smallest edge, connect components; continue — components merge over time until one component.

Prim's Algorithm

Idea: Start from any vertex; repeatedly add the cheapest edge connecting the growing tree to a new vertex (like Dijkstra but for MST).

Pseudocode (using min-heap / priority queue):

```
Prim(G, r):
  for each v: key[v] = ∞, parent[v] = NIL
  key[r] = 0
  Q = all vertices with key
  while Q not empty:
    u = ExtractMin(Q)
    for each v adjacent to u:
      if v in Q and weight(u,v) < key[v]:
        parent[v] = u
        key[v] = weight(u,v)
  return parent[] // edges form MST
```

Variants:

- Dense graph: adjacency matrix + $O(V^2)$ implementation is simple and efficient for large V .
- Sparse graph: adjacency list + binary heap $\rightarrow O(E \log V)$; with Fibonacci heap $\rightarrow O(E + V \log V)$.

Complexity:

- Binary heap: $O(E \log V)$.
- Fibonacci heap: $O(E + V \log V)$.
- Adj matrix (no heap): $O(V^2)$.

When to use Kruskal vs Prim:

- Kruskal is often simpler when E

You said:

wait a minute , recreate but using c language , or try to make the step wise algorithm

ChatGPT said:

Great — I will **recreate everything**, but now **purely in C-style**, with:

- ✓ Step-wise algorithms
- ✓ C-language style functions

- ✓ Visual explanations
- ✓ Time & space complexity for each

I'll cover the topics in this order:

1. **Greedy Strategy – in C-oriented explanation**
 2. **Huffman Coding – stepwise + C implementation**
 3. **Minimum Spanning Trees**
 - **Kruskal's Algorithm – stepwise + C implementation**
 - **Prim's Algorithm – stepwise + C implementation**
 4. **Single-Source Shortest Path**
 - **Dijkstra's Algorithm – stepwise + C implementation**
 - **Bellman–Ford Algorithm – stepwise + C implementation**
 5. **All-Pairs Shortest Path – Floyd–Warshall Algorithm**
 6. **Topological Sort – stepwise + C implementation**
-

★ 1) GREEDY STRATEGY (with C-style)

Meaning

A greedy algorithm *always picks the locally optimal choice* hoping it leads to a global optimum.

GREEDY STRATEGY – STEPWISE

1. **Define the optimization goal**
e.g., minimize cost, maximize profit.
 2. **Define the choice set**
e.g., edges, frequencies, intervals.
 3. **Greedy-choice property**
The choice that looks best now must also appear in some optimal solution.
 4. **Optimal substructure**
After choosing, the remaining part must also be optimally solvable.
 5. **Make the choice**
Usually implemented with:
 - Sorting
 - Min-heap / priority queue
 - Selecting smallest/largest element
-

C-style Template

```
// Greedy Template
void greedySolve(Element arr[], int n) {
    // Step 1: Preprocessing (sort or push to heap)
    prepare(arr, n);

    Solution S = emptySolution();

    for (int i = 0; i < n; i++) {
        if (isFeasible(S, arr[i])) {
            S = include(S, arr[i]);
        }
    }

    printSolution(S);
}
```

★ 2) HUFFMAN CODING (C version)

Visualization

Think of each character as a leaf.

(A:5) (B:9) (C:12) (D:13) (E:16) (F:45)
Put all into min-heap.

Step-by-step merge:

1. Take A(5), B(9) → Node(14)
 2. Take C(12), Node(14) → Node(26)
 3. Take D(13), E(16) → Node(29)
 4. Take Node(26), Node(29) → Node(55)
 5. Take F(45), Node(55) → ROOT(100)
-

HUFFMAN – ALGORITHM (stepwise)

1. Create a **min-heap** of all characters with frequencies.
2. While size > 1:
 - a. Remove two minimum-frequency nodes
 - b. Create a new node with frequency = sum
 - c. Insert back into heap
3. Final node = root of Huffman Tree
4. Assign
 - left = 0

- o right = 1

C IMPLEMENTATION (compact)

```
#include <stdio.h>
#include <stdlib.h>

typedef struct Node {
    char data;
    int freq;
    struct Node *left, *right;
} Node;

// Create a new node
Node* newNode(char d, int f) {
    Node* n = malloc(sizeof(Node));
    n->data = d;
    n->freq = f;
    n->left = n->right = NULL;
    return n;
}
```

Min Heap for Huffman

```
typedef struct {
    int size;
    int capacity;
    Node** array;
} MinHeap;

// swap
void swap(Node** a, Node** b) {
    Node* t = *a; *a = *b; *b = t;
}
```

Build Huffman Tree

```
Node* buildHuffman(char data[], int freq[], int n) {
    MinHeap* heap = buildMinHeap(data, freq, n);

    while (heap->size > 1) {
        Node *l = extractMin(heap);
        Node *r = extractMin(heap);

        Node* top = newNode('$', l->freq + r->freq);
        top->left = l;
        top->right = r;

        insertMinHeap(heap, top);
    }
}
```

```
    return extractMin(heap); // root
}
```

Time Complexity

- Heap operations: $O(n \log n)$
- Tree building: $O(n)$ merges
- **Total: $O(n \log n)$**

Space Complexity

- Tree nodes: $O(n)$
-

★ 3) MINIMUM SPANNING TREES (MST)

A) KRUSKAL'S MST

Visualization

Imagine edges sorted:

```
1 -- 2 (2)
2 -- 3 (3)
1 -- 3 (5)
3 -- 4 (7)
2 -- 4 (9)
```

Pick smallest that **doesn't form a cycle**.

KRUSKAL – STEPWISE ALGORITHM

1. Sort edges by increasing weight
2. Initialize Disjoint-Set (Union-Find)
3. For each edge (u, v) :

- If Find(u) != Find(v)
 - include in MST
 - Union(u, v)
 - 4. Stop after V-1 edges chosen
-

KRUSKAL – C FUNCTION

```
typedef struct {
    int src, dest, weight;
} Edge;

int find(int parent[], int x) {
    if (parent[x] != x)
        parent[x] = find(parent, parent[x]);
    return parent[x];
}

void Union(int parent[], int rank[], int x, int y) {
    x = find(parent, x);
    y = find(parent, y);

    if (rank[x] < rank[y]) parent[x] = y;
    else if (rank[x] > rank[y]) parent[y] = x;
    else { parent[y] = x; rank[x]++; }
}

void Kruskal(Edge edges[], int E, int V) {
    // Step 1: Sort edges by weight
    sort(edges, E);

    int parent[V], rank[V];
    for (int i = 0; i < V; i++)
        parent[i] = i, rank[i] = 0;

    int count = 0;

    for (int i = 0; i < E && count < V-1; i++) {
        int u = edges[i].src;
        int v = edges[i].dest;

        if (find(parent, u) != find(parent, v)) {
            printf("%d -- %d = %d\n", u, v, edges[i].weight);
            Union(parent, rank, u, v);
            count++;
        }
    }
}
```

Time Complexity

- Sorting edges $\rightarrow O(E \log E)$
- Union-Find $\rightarrow$ almost $O(E)$
- Total $\rightarrow O(E \log E)$

Space Complexity

- Parent + rank arrays $\rightarrow O(V)$

B) PRIM'S MST

PRIM – STEPWISE ALGORITHM

1. Choose a starting vertex
2. Set $\text{key}[\text{source}] = 0$, others = INF
3. Use min-priority queue
4. Repeatedly extract vertex u with minimum key
5. Relax edges (u, v) :
 - if $\text{weight} < \text{key}[v]$, update v
6. Continue until all vertices added

PRIM – C IMPLEMENTATION (Adjacency Matrix)

```
#define INF 99999

int minKey(int key[], int mstSet[], int V) {
    int min = INF, idx = -1;
    for (int v = 0; v < V; v++)
        if (!mstSet[v] && key[v] < min)
            min = key[v], idx = v;
    return idx;
}

void Prim(int graph[10][10], int V) {
    int parent[V];
    int key[V];
    int mstSet[V];

    for (int i = 0; i < V; i++)
        key[i] = INF, mstSet[i] = 0;

    key[0] = 0; parent[0] = -1;
```

```

for (int i = 0; i < V - 1; i++) {
    int u = minKey(key, mstSet, V);
    mstSet[u] = 1;

    for (int v = 0; v < V; v++) {
        if (graph[u][v] && !mstSet[v] && graph[u][v] < key[v]) {
            parent[v] = u;
            key[v] = graph[u][v];
        }
    }
}

for (int i = 1; i < V; i++)
    printf("%d -- %d = %d\n", parent[i], i, graph[i][parent[i]]);
}

```

Time Complexity

- Using adjacency matrix: $O(V^2)$
- Using heap: $O(E \log V)$

Space Complexity

- Key, parent, mstSet $\rightarrow O(V)$
-

★ 4) SHORTEST PATH ALGORITHMS

A) DIJKSTRA'S ALGORITHM – no negative edges

DIJKSTRA – STEPWISE

1. $\text{dist}[\text{source}] = 0$
2. All other $\text{dist}[] = \text{INF}$
3. Use min-priority queue (or $O(V^2)$ array)
4. Extract vertex u with smallest dist

5. Relax edges:
if $\text{dist}[v] > \text{dist}[u] + w(u, v)$
 $\text{dist}[v] = \text{dist}[u] + w(u, v)$
-

DIJKSTRA – C IMPLEMENTATION

```
#define INF 99999

int minDistance(int dist[], int visited[], int V) {
    int min = INF, idx = -1;
    for (int i = 0; i < V; i++)
        if (!visited[i] && dist[i] < min)
            min = dist[i], idx = i;
    return idx;
}

void Dijkstra(int graph[10][10], int V, int src) {
    int dist[V], visited[V];

    for (int i = 0; i < V; i++)
        dist[i] = INF, visited[i] = 0;

    dist[src] = 0;

    for (int c = 0; c < V - 1; c++) {
        int u = minDistance(dist, visited, V);
        visited[u] = 1;

        for (int v = 0; v < V; v++)
            if (graph[u][v] && !visited[v] && dist[u] + graph[u][v] < dist[v])
                dist[v] = dist[u] + graph[u][v];
    }

    for (int i = 0; i < V; i++)
        printf("dist[%d] = %d\n", i, dist[i]);
}
```

Complexity

- $O(V^2)$ using matrix
 - $O(E \log V)$ using heap
-

B) BELLMAN–FORD – handles negative weights

BELLMAN-FORD – STEPWISE

1. $\text{dist}[\text{src}] = 0$; others = INF
 2. Repeat $(V-1)$ times:
For every edge (u, v) :
relax $\text{dist}[v] = \min(\text{dist}[v], \text{dist}[u] + w)$
 3. Check negative cycle (one more relaxation)
-

BELLMAN-FORD – C CODE

```
void BellmanFord(Edge edges[], int V, int E, int src) {
    int dist[V];
    for (int i = 0; i < V; i++)
        dist[i] = INF;
    dist[src] = 0;

    for (int i = 1; i <= V-1; i++)
        for (int j = 0; j < E; j++) {
            int u = edges[j].src;
            int v = edges[j].dest;
            int w = edges[j].weight;

            if (dist[u] != INF && dist[u] + w < dist[v])
                dist[v] = dist[u] + w;
        }

    // Cycle check
    for (int j = 0; j < E; j++) {
        int u = edges[j].src;
        int v = edges[j].dest;
        if (dist[u] + edges[j].weight < dist[v])
            printf("Negative Cycle Found!\n");
    }
}
```

Time Complexity

- $O(V \times E)$
-

★ 5) ALL-PAIRS SHORTEST PATH

FLOYD–WARSHALL ALGORITHM

STEPWISE

1. $\text{dist}[i][j] = \text{weight}(i, j)$; 0 if $i=j$; INF if no edge
 2. For $k = 0$ to $V-1$
 3. For $i = 0$ to $V-1$
 4. For $j = 0$ to $V-1$
 - o $\text{dist}[i][j] = \min(\text{dist}[i][j], \text{dist}[i][k] + \text{dist}[k][j])$
-

C CODE

```
void FloydWarshall(int dist[][10], int V) {
    for (int k = 0; k < V; k++)
        for (int i = 0; i < V; i++)
            for (int j = 0; j < V; j++)
                if (dist[i][k] + dist[k][j] < dist[i][j])
                    dist[i][j] = dist[i][k] + dist[k][j];
}
```

Time Complexity

- $O(V^3)$
Space: $O(V^2)$
-

★ 6) TOPOLOGICAL SORT (Directed Acyclic Graph)

TOPOLOGICAL SORT – STEPWISE

1. Compute in-degree of all vertices
2. Put all in-degree 0 vertices into queue
3. Remove one vertex at a time

4. Reduce in-degree of neighbors
 5. Any neighbor whose in-degree becomes 0 $\rightarrow$ enqueue
 6. Continue until queue empty
 7. If processed vertices $< V \rightarrow$ cycle exists
-

C CODE

```
void topoSort(int graph[10][10], int V) {
    int indeg[10] = {0};

    for (int i = 0; i < V; i++)
        for (int j = 0; j < V; j++)
            if (graph[i][j])
                indeg[j]++;

    int queue[10], front = 0, rear = 0;

    for (int i = 0; i < V; i++)
        if (indeg[i] == 0)
            queue[rear++] = i;

    while (front != rear) {
        int u = queue[front++];

        printf("%d ", u);

        for (int v = 0; v < V; v++)
            if (graph[u][v]) {
                indeg[v]--;
                if (indeg[v] == 0)
                    queue[rear++] = v;
            }
    }
}
```

✓ COMPLETED

I have now covered **EVERY** topic in your Unit—
with:

- ✓ full step-wise algorithm
- ✓ C-style code for each
- ✓ visualization
- ✓ time & space complexity